AMELIA REGINA - 2C2230019 TUGAS 7 STRUKTUR DATA

PRACTICE

MEMBUAT NODE

```
class Node:
    def __init__(self, data):
        self.data = data

node1 = Node(10)

print(node1.data)
```

```
10
```

MEMBUAT LINKED LIST SEDERHANA

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

node1 = Node(10)
node2 = Node(20)

node1.next = node2

print(node1.next.data)
```

```
20
```

Traversal Linked List

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

node1 = Node(10)
node2 = Node(20)
node3 = Node(30)
node4 = Node(40)

node1.next = node2
node2.next = node3
node3.next = node4

temp = node1

while temp:
    print(temp.data, end=" -> ")
    temp = temp.next
```

```
print("None")
```

```
10 -> 20 -> 30 -> 40 -> None
```

Insert Node

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(20)
head.next = Node(30)

# tambah di awal
baru_awal = Node(10)
baru_awal.next = head
head = baru_awal

# tambah di akhir
baru_akhir = Node(40)

temp = head
while temp.next:
    temp = temp.next

temp.next = baru_akhir

temp = head
while temp:
    print(temp.data, end=" -> ")
    temp = temp.next

print("None")
```

```
10 -> 20 -> 30 -> 40 -> None
```

Delete Node

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
head.next = Node(20)
head.next.next = Node(30)

hapus = 20

temp = head

while temp.next:
    if temp.next.data == hapus:
        temp.next = temp.next.next
        break
    temp = temp.next
```

```
temp = head
while temp:
    print(temp.data, end=" -> ")
    temp = temp.next

print("None")
```

```
10 -> 30 -> None
```

Search Data

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
head.next = Node(20)
head.next.next = Node(30)

cari = 20
ketemu = False

temp = head

while temp:
    if temp.data == cari:
        ketemu = True
        break
    temp = temp.next

if ketemu:
    print("Data ditemukan")
else:
    print("Data tidak ditemukan")
```

```
Data ditemukan
```

Doubly Linked List

```
class Node:
    def __init__(self, data):
        self.data = data
        self.prev = None
        self.next = None

node1 = Node(10)
node2 = Node(20)
node3 = Node(30)

node1.next = node2
node2.prev = node1

node2.next = node3
node3.prev = node2

temp = node1
```

```
    while temp:
        print(temp.data, end=" <-> ")
        temp = temp.next

    print("None")
```

```
10 <-> 20 <-> 30 <-> None
```

Circular Linked List

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

node1 = Node(10)
node2 = Node(20)
node3 = Node(30)

node1.next = node2
node2.next = node3
node3.next = node1

temp = node1

for i in range(6):
    print(temp.data, end=" -> ")
    temp = temp.next

print("...")
```

```
10 -> 20 -> 30 -> 10 -> 20 -> 30 -> ...
```

TUGAS PRAKTIKUM

LEVEL MUDAH

TUGAS 1 - Buat Node sederhana dan tampilkan datanya

```
class Node:
    def __init__(self, data):
        self.data = data

node1 = Node('kucing')

print("Data node =", node1.data)
```

```
Data node = kucing
```

TUGAS 2 - Buat Linked List berisi 5 angka

```
class Node:
    def __init__(self, data):
        self.data = data
```

```
        self.next = None

n1 = Node(10)
n2 = Node(20)
n3 = Node(30)
n4 = Node(40)
n5 = Node(50)

n1.next = n2
n2.next = n3
n3.next = n4
n4.next = n5

head = n1
```

TUGAS 3 - Traversal seluruh node

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

n1 = Node("Kucing")
n2 = Node("Anjing")
n3 = Node("Kelinci")
n4 = Node("Sapi")
n5 = Node("Kambing")

n1.next = n2
n2.next = n3
n3.next = n4
n4.next = n5

head = n1

current = head

while current:
    print(current.data)
    current = current.next
```

```
Kucing
Anjing
Kelinci
Sapi
Kambing
```

TUGAS 4 - Tambahkan node di awal Linked List

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

n1 = Node("Photograph")
n2 = Node("Perfect")

n1.next = n2
```

```
head = n1

baru = Node("Shape of You")
baru.next = head
head = baru

current = head

while current:
    print(current.data)
    current = current.next
```

```
Shape of You
Photograph
Perfect
```

LEVEL MENENGAH

Tugas 5 - Tambahkan Node di Akhir Linked List

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node('jeruk')
head.next = Node('apel')
head.next.next = Node('mangga')

baru = Node('melon')

temp = head
while temp.next:
    temp = temp.next

temp.next = baru

temp = head
while temp:
    print(temp.data, end=" -> ")
    temp = temp.next

print("None")
```

```
jeruk -> apel -> mangga -> melon -> None
```

Tugas 6 - Hapus Node Tertentu

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node('ferari')
head.next = Node('mercedess')
head.next.next = Node('redbull')
```

```
hapus = 'ferari'

temp = head

while temp.next:
    if temp.next.data == hapus:
        temp.next = temp.next.next
        break
    temp = temp.next

temp = head
while temp:
    print(temp.data, end=" -> ")
    temp = temp.next

print("None")
```

```
ferari -> mercedess -> redbull -> None
```

Tugas 7 - Cari Data Tertentu pada Linked List

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
head.next = Node(20)
head.next.next = Node(30)

cari = 20
ketemu = False

temp = head

while temp:
    if temp.data == cari:
        ketemu = True
        break
    temp = temp.next

if ketemu:
    print("Data ditemukan")
else:
    print("Data tidak ditemukan")
```

```
Data ditemukan
```

Tugas 8 - Hitung Jumlah Node

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(11)
head.next = Node(13)
head.next.next = Node(32)
head.next.next.next = Node(40)
```

```python
jumlah = 0

temp = head

while temp:
    jumlah += 1
    temp = temp.next

print("Jumlah node =", jumlah)
```

```
Jumlah node = 4
```